Computational Plasma Physics: Portfolio Assignment 1

December 2025

3 Normalisation

The time and length scale provided by the script were the thermal hydrogen gyro-period, τ_H, and the gyro-radius, ρ_H.

$$\tau_H = \frac{m_H}{q_H B_r}, \quad \rho_H = \frac{m_H v_{th,H}}{q_H B_r} = v_{th,H}\tau_H \tag{1}$$

These are the normalisation equations needed.

$$\hat{B} = \frac{\vec{B}}{B_r}, \quad \hat{v} = \frac{\vec{v}}{v_{th,H}}, \quad \hat{x} = \frac{\vec{x}}{\rho_H}, \quad \hat{E} = \frac{\vec{E}}{B_r v_{th,H}}, \quad \hat{t} = \frac{t}{\tau_H}, \quad \hat{m} = \frac{m}{m_H}, \quad \hat{q} = \frac{q}{q_H} \tag{2}$$

The equation of motion for a charged particle within an electric and magnetic field is as follows.

$$\frac{m}{q}\frac{d^2\vec{x}}{dt^2} = \vec{E} + \vec{v} \times \vec{B} \tag{3}$$

Splitting this into each of the Cartesian components gives the following.

$$\frac{m}{q}\frac{d^2x}{dt^2} = E_x + v_y B_z - v_z B_y, \quad \frac{m}{q}\frac{d^2y}{dt^2} = E_y - v_x B_z + v_z B_x, \quad \frac{m}{q}\frac{d^2z}{dt^2} = E_z + v_x B_y - v_y B_x \tag{4}$$

Applying the normalisation equations and the time and length scales to these equations gives the following equations.

$$\frac{\hat{m}}{\hat{q}}\frac{m_H}{q_H}\frac{d^2\hat{x}}{d\hat{t}^2}\frac{q_H B_r v_{th,H}}{m_H} = B_r V_{th,H}(\hat{E_x} + \hat{v_y}\hat{B_z} - \hat{v_z}\hat{B_y}) \tag{5}$$

$$\frac{\hat{m}}{\hat{q}}\frac{m_H}{q_H}\frac{d^2\hat{y}}{d\hat{t}^2}\frac{q_H B_r v_{th,H}}{m_H} = B_r V_{th,H}(\hat{E_y} - \hat{v_x}\hat{B_z} + \hat{v_z}\hat{B_x}) \tag{6}$$

$$\frac{\hat{m}}{\hat{q}}\frac{m_H}{q_H}\frac{d^2\hat{z}}{d\hat{t}^2}\frac{q_H B_r v_{th,H}}{m_H} = B_r V_{th,H}(\hat{E_z} + \hat{v_x}\hat{B_y} - \hat{v_y}\hat{B_x}) \tag{7}$$

By cancelling out values, the following equations can be achieved.

$$\frac{d^2\hat{x}}{d\hat{t}^2} = \frac{\hat{q}}{\hat{m}}(\hat{E_x} + \hat{v_y}\hat{B_z} - \hat{v_z}\hat{B_y}) \tag{8}$$

$$\frac{d^2\hat{y}}{d\hat{t}^2} = \frac{\hat{q}}{\hat{m}}(\hat{E_y} - \hat{v_x}\hat{B_z} + \hat{v_z}\hat{B_x}) \tag{9}$$

$$\frac{d^2\hat{z}}{d\hat{t}^2} = \frac{\hat{q}}{\hat{m}}(\hat{E_z} + \hat{v_x}\hat{B_y} - \hat{v_y}\hat{B_x}) \tag{10}$$

4 Implementation

The script provides us with the information: $B_x = 0$, $B_y = 0$, $E_z = 0$, $v_z = 0$. By plugging these into equations (8), (9), and (10) we get the following:

$$\frac{d^2\hat{x}}{d\hat{t}^2} = \frac{\hat{q}}{\hat{m}}(\hat{E_x} + \hat{v_y}\hat{B_z}) \tag{11}$$

$$\frac{d^2\hat{y}}{d\hat{t}^2} = \frac{\hat{q}}{\hat{m}}(\hat{E_y} - \hat{v_x}\hat{B_z}) \tag{12}$$

$$\frac{d^2\hat{z}}{d\hat{t}^2} = 0 \tag{13}$$

5 Application

Task 1: Particle motion explained

The particles with a positive charge: 0, 2, and 3 each move in a clockwise direction, whereas the particle with a negative charge, 1, moves in an anti-clockwise direction.

The particles with a mass of 0.1, 2 and 3, each experience more acceleration from moving through the B field than the particles with mass 1, 0 and 1, causing them to move in tighter spirals.

When there is no electric field, the positively charged particles: 0, 2, and 3 each experience a drift in the negative $\hat{y}$ direction, while the negatively charged particle, 1, experiences a drift in the positive $\hat{y}$ direction. This drift is perpendicular to the gradient of the B field and is a result of Grad B drift.

When an electric field is present, all the particles experience a drift in the positive $\hat{x}$ direction in addition to the grad B drift. This drift is perpendicular to both the magnetic field in the positive $\hat{z}$ direction and the electric field in the positive $\hat{y}$ direction. It is a result of ExB drift.

Task 2: Calculate kinetic energy

First, an equation for the normalised kinetic energy must be derived. ϵ represents the kinetic energy to avoid confusion with electric field.

$$\hat{m} = \frac{m}{m_H}, \quad \hat{v} = \frac{v}{v_{th,H}}, \quad \hat{\epsilon} = \frac{2\epsilon}{m_H v_{th,H}^2}. \tag{14}$$

$$\epsilon = \frac{mv^2}{2} \tag{15}$$

Substituting in the normalised values gives:

$$\frac{\hat{\epsilon} m_H v_{th,H}^2}{2} = \frac{\hat{m}\hat{v}^2 m_H v_{th,H}^2}{2} \tag{16}$$

This simplifies to:

$$\hat{\epsilon} = \hat{m}\hat{v}^2 \tag{17}$$

The kinetic energy plot can be produced using 'python3 ChargedParticleMotion.py -s 3'.

The average kinetic energy of all particles increased by 0.0112 +/- 0.0016 over 500 gyro-periods. This is a ratio of 1.01 for the particles of mass 1 and a ratio of 1.1 for the particles of mass 0.1. Ideally we should expect a ratio of 1 because no energy is being added or lost from the system. There also arises a small oscillation (0.001 energy units) in the energy. I think these small variations are due to instabilities in the Runge-Kutta 4 method. However, this uncertainty is acceptable for this use case.

Task 3: Modify code to introduce E field

The position plot with electric field can be produced using 'python3 ChargedParticleMotion.py -s 4' and the kinetic energy plot with electric field can be produced using 'python3 ChargedParticleMotion.py -s 5'.

Task 4: Describe change in KE conservation

Adding an electric field causes the kinetic energy of the particles to change from (mostly) constant to varying sinusoidally around the initial energy. This is caused by the electric field periodically accelerating and decelerating the particles as they orbit due to the magnetic field.

When the electric field is added the particles lose energy over time instead of staying constant (simulation needs running for longer to better see this in the lower mass particles.)

When no electric field is present the kinetic energies of the particles remain roughly constant. Adding an electric field varies the kinetic energy of the particles in two ways:

- On short time scales the kinetic energy oscillates about the initial energy. This is caused by the particles being periodically accelerated and decelerated by the electric field as they move with and against it due to its gyro motion due to the magnetic field.
- On larger time scales the kinetic energy decreases over time (simulation needs running for longer to better see this in the lower mass particles.) The particle is moving through an inhomogeneous magnetic field which decreases in strength as $\hat{x}$ increases, the gyro-period therefore decreases as $\hat{x}$ increases. Because of this, the particles spend more time in the portion of their gyro motion when they are being decelerated by the electric field. This results in a net deceleration and net loss of kinetic energy.